

SuryaSugandh

A hybrid solar + 230 V grid drying and packaging machine for home-based agarbatti (incense stick) manufacturing.

Problem Statement: **PS 22:** Smart, solar-powered drying and compact packaging system to support home-based agarbatti manufacturing by rural women artisans.

Problem

Home-based agarbatti makers dry sticks in the sun or ambient heat, which takes about **9 hours** and depends on the weather. Night and rainy-day drying is difficult, monitoring is manual, output quality varies, and packing is done by hand.

Our Solution

SuryaSugandh is a compact machine that combines **forming, drying, fragrance treatment, weighing and sealing** in one unit. It runs on a **hybrid power system** (100 Wp solar panel + 230 V grid backup), so it works in sunlight, at night and in bad weather.

Key features

- Drying time cut from **9 h to 4–5 h**
- Temperature and humidity controlled by an **ESP32**
- Solar-first power with automatic grid backup
- Semi-automatic packaging: weigh/count, feed, seal (50 g packets)
- Low hardware cost: **Rs 18,200**

Slides

Technical Approach	Feasibility & Viability
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Technical Approach

1. Production flow

Forming & Coating → Drying → Fragrance → Weigh
/ Count → Heat Sealing → Packet Output

Stage	What happens	Rating
Forming & coating	Forms and coats raw sticks (intermittent)	150 W motor
Drying	Thermostat-controlled hot air with moisture removal	500 W heater + 80 W fans, 4–5 h

Stage	What happens	Rating
Fragrance	Controlled fragrance application	20 W pump
Weigh / count	Load cell or preset count, 50 g target packet (100 sticks)	–
Heat sealing	Pouch positioned and sealed, one at a time	100 W sealer
Output	Sealed packet moves to collection area	40 packets per 2 kg batch

2. Hybrid power arrangement

Solar panel (100 Wp) → Charge controller (40 A) → 24 V heater controller → 500 W heater
 Solar panel → Charge controller → Battery (12 V, 75 Ah) → ESP32 / sensors / fans
 230 V grid → 24 V AC-DC supply (600 W) → 24 V heater

- Solar runs the heater during useful sunlight; the battery supports the controller, sensors, display and low-power loads.
- The grid feeds the heater when solar is insufficient or at night.
- Changeover switch + MCB handle source selection and protection.

3. Smart control (ESP32)

Sensors (Temperature · Humidity · PIR) → ESP32 →
Relays (heater, fans, pump) + Display +
Buzzer/LEDs

Components and Cost

#	Component	Rating	Cost (Rs)	Why we use it
1	Heating element	500 W	1,000	Dries sticks faster; thermostat holds temperature
2	DC circulation/exhaust fans	80 W	500	Air circulation and moisture removal
3	Agarbatti forming motor	150 W	1,000	Forms and coats raw sticks
4	Fragrance pump	20 W	400	Controlled fragrance application

#	Component	Rating	Cost (Rs)	Why we use it
5	Packaging / sealing unit	100 W	700	Weighing and sealing packets
6	ESP32 controller	5 W	450	Main control; reads sensors, switches loads
7	Temp / RH / PIR sensors	5 W	500	Monitoring and safety
8	Display	3 W	300	Shows status and readings
9	Buzzer, relays, LEDs	5 W	250	Load switching, alerts, indication
10	Solar panel	100 Wp	3,500	Primary solar source

#	Component	Rating	Cost (Rs)	Why we use it
11	Solar charge controller	40 A	900	Battery protection and DC distribution
12	24 V DC heater controller	–	800	Regulates solar power to the heater
13	24 V AC-DC supply	600 W	1,500	Grid backup for the heater
14	Battery	12 V, 75 Ah	4,500	Backup for electronics and low-power loads
15	Inverter	1 kVA	1,200	Small AC backup loads
16	Changeover, MCB, wiring	–	700	Source selection and protection
	TOTAL		18,200	

Energy Consumption

Load	Rated power	Daily ON time	Daily energy
Heating element	500 W	6 h	3,000 Wh
Fans	80 W	10 h	800 Wh
Forming motor	150 W	4 h	600 Wh
Fragrance pump	20 W	1 h	20 Wh
Packaging / sealing	100 W	1 h	100 Wh
ESP32	5 W	10 h	50 Wh
Sensors	5 W	10 h	50 Wh
Display	3 W	10 h	30 Wh
Buzzer / relays / LEDs	5 W	10 h	50 Wh
TOTAL			4,700 Wh = 4.7 kWh/day

Monthly: **141 kWh** · Yearly: **1,715.5 kWh**

Electricity cost: grid only vs hybrid (Rs 8/unit)

Mode	Units/day	Units/month	Bill/month	Bill/year
Grid only	4.7	141	Rs 1,128	Rs 13,724
Hybrid solar + grid	4.4	132	Rs 1,056	Rs 12,848

Feasibility and Viability

Feasibility

- **Technical:** standard low-cost parts (ESP32, 40 A charge controller, 24 V supplies); plugs into a normal 230 V socket.
- **Operational:** grid backup allows night and cloudy-day use; temperature + humidity control gives consistent output.
- **Economic:** 4.7 kWh/day; hybrid bill Rs 1,056/month vs Rs 1,128 grid-only; operating cost Rs 185 vs Rs 290 per day.

Viability (8-hour working day, 2 kg batch, 0.5 g stick, Rs 20 per 50 g packet)

	Traditional pedal	Proposed machine
Drying time / batch	9 h	4.5 h (4–5 h)
Batches (2 kg) in 8 h	0.89	1.78
Sticks per day	≈ 3,556	≈ 7,111
50 g packets per day	≈ 35.6	≈ 71.1
Sales / day	Rs 711	Rs 1,422
Operating cost / day	Rs 290	Rs 185
Profit / day	Rs 421	Rs 1,237
Profit / month (30 days)	Rs 12,630	Rs 37,110
Profit / year (365 days)	Rs 153,665	Rs 451,505

- Drying-stage throughput: **≈ 2×** the traditional method
- Extra profit: **Rs 816/day**, so hardware payback is about **22 working days** (Rs 18,200 ÷ Rs 816)

Repository Structure

```
SuryaSugandh/  
├── README.md  
├── docs/      # slides, project PDF,  
calculations  
├── images/    # prototype photos, diagrams  
├── hardware/  # component list, circuit  
diagram  
└── code/      # ESP32 code
```

Team

- **Team name:** *add here*
- **Members:** *add names here*
- **College:** *add here*

Links

- Demo video: *add link*
- Presentation: `docs/`

*Built for PS 22 · Solar-powered drying and packaging for
rural women artisans*